SENTINEL

SENTINEL: AI-Powered GPU Infrastructure Management Platform

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Table Of Content

- TITLE SLIDE
- PROBLEM STATEMENT
- INDUSTRY MOTIVATION
- EXISTING SYTEM
- PROBLEM DEFINITION
- PROPOSED SOLUTION
- SYSTEM ARCHITECTURE
- TECHNOLOGY STACK
- FUTURE INNOVATIONS
- IEEE REFERENCES

Presented By Group 23

INTRODUCTION

- Rapid advancements in Artificial Intelligence (AI), Machine Learning (ML), and High-Performance Computing (HPC) have significantly increased the deployment of large-scale GPU clusters in modern data centers.
- Existing GPU monitoring tools primarily provide basic hardware metrics and threshold-based alerts, making it difficult to proactively identify performance degradation and potential hardware failures.
- The proposed Enterprise AIOps Platform for Intelligent GPU Infrastructure Management leverages AI to continuously monitor GPU telemetry, detect anomalies, assess GPU health, and predict potential failures in real time.

PROBLEM STATEMENT:

AI-OPS-BASED INTELLIGENT GPU CLUSTER OBSERVABILITY PLATFORM FOR PREDICTIVE MAINTENANCE AND RESOURCE OPTIMIZATION

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INDUSTRY MOTIVATION

The AI Revolution is Driving Massive GPU Infrastructure Growth

The rapid adoption of Generative AI, Large Language Models (LLMs), and High-Performance Computing (HPC) has significantly increased the demand for large-scale GPU infrastructure. Organizations worldwide are investing billions of dollars in AI data centers to support model training and inference, making efficient GPU management a critical operational challenge.

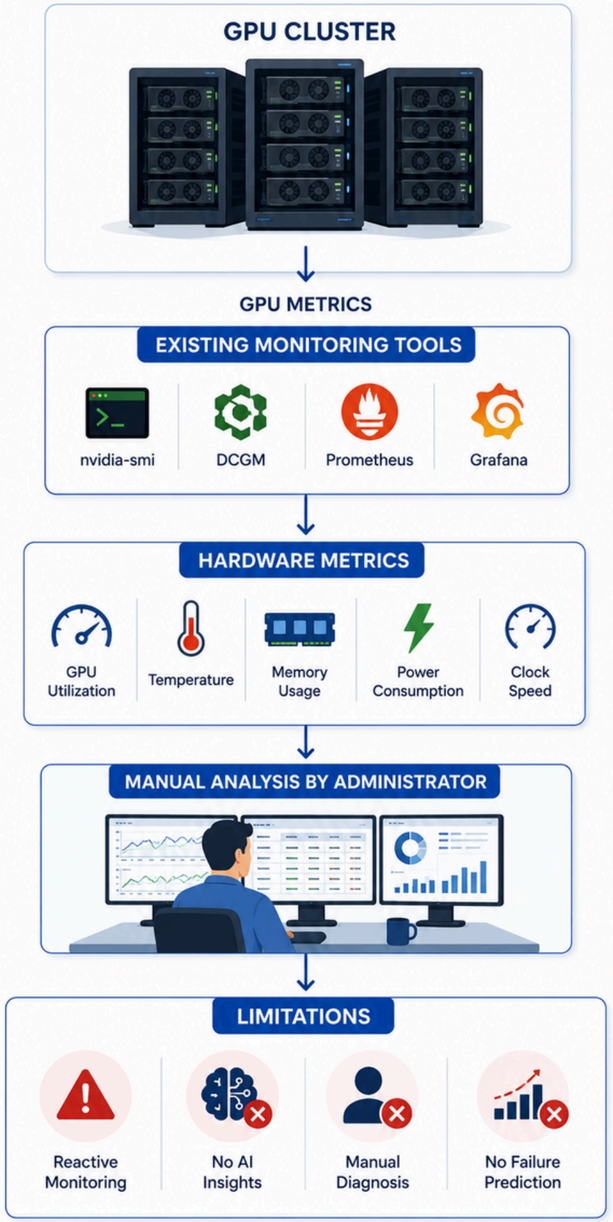
INDUSTRY DRIVERS

- **NVIDIA**
  - World's leading provider of AI GPUs powering modern data centers.
  - Developing next-generation AI infrastructure with DGX and Blackwell platforms.
- **OpenAI**
  - Operates massive GPU clusters to train and deploy Large Language Models.
  - Requires highly reliable GPU infrastructure to ensure continuous AI services.
- **Google**
  - Uses thousands of GPUs and TPUs across Google Cloud to support Gemini and enterprise AI workloads.
  - Focuses on intelligent resource management and infrastructure optimization.

EXISTING SYSTEMS

Limitations of Existing GPU Monitoring Solutions

- Existing GPU monitoring relies on tools such as **nvidia-smi, DCGM, Prometheus, and Grafana** for real-time telemetry and performance monitoring.
- These platforms monitor **GPU utilization, memory usage, temperature, power consumption, and hardware status** using dashboards and threshold-based alerts.
- Most existing solutions are **reactive**, requiring manual analysis after performance issues or failures occur.
- They **lack AI-driven capabilities** such as anomaly detection, GPU Health Scoring, failure prediction, and intelligent maintenance recommendations.
- Growing GPU infrastructures demand an **AI-powered observability** platform capable of proactive monitoring and predictive maintenance.



PROBLEM DEFINITION

1. Manual GPU Monitoring

Existing GPU infrastructures rely heavily on manual monitoring, making it difficult to efficiently manage hundreds or thousands of GPUs.

2. Reactive Fault Detection

Current monitoring tools identify issues only after they occur, leading to delayed response and increased system downtime.

3. Lack of AI Intelligence

Traditional solutions do not provide anomaly detection, failure prediction, or intelligent maintenance recommendations.

4. Need for Proactive Monitoring

There is a growing need for an AI-powered platform capable of real-time monitoring and predictive analytics altogether

SO WHAT EXACTLY ARE WE BUILDING?

PROPOSED SOLUTION

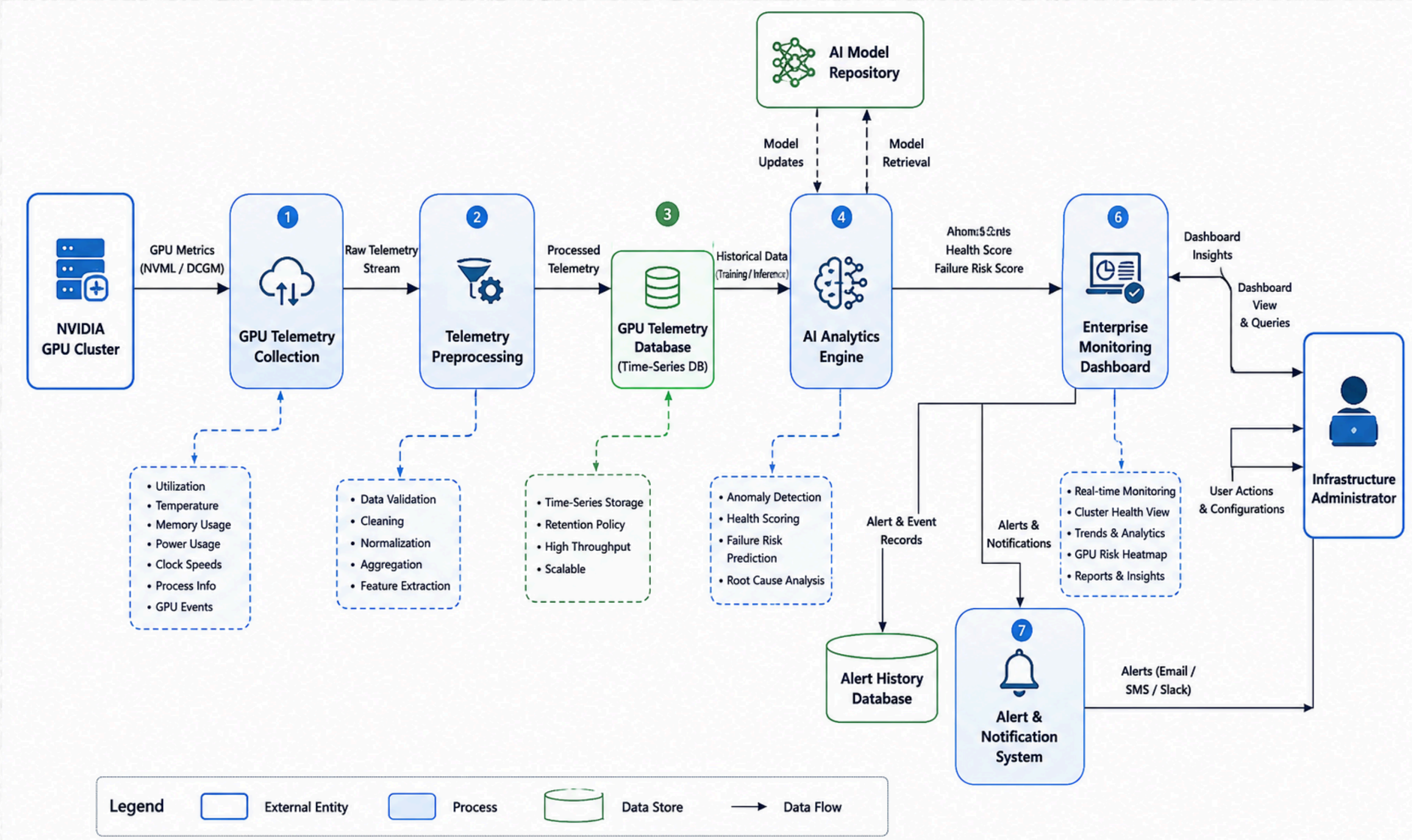
Introducing **Sentinel** – An Enterprise AIOps Platform for Intelligent GPU Infrastructure Management

- SENTINEL is an **AI-powered Enterprise AIOps platform** designed to provide intelligent monitoring, anomaly detection, and predictive maintenance for **large-scale GPU infrastructures** powering modern AI data centers. Unlike conventional GPU monitoring tools that rely on threshold-based alerts and manual analysis, **SENTINEL** continuously analyzes **real-time GPU telemetry** using Machine Learning to detect abnormal behavior, assess GPU health, estimate failure risks, and provide **proactive maintenance** recommendations.
- The platform collects **real-time GPU telemetry** using NVIDIA monitoring interfaces and leverages AI to detect anomalies, **generate GPU Health Scores**, and **estimate Failure Risk Scores** across the cluster.

KEY CAPABALITIES

- REAL-TIME GPU TELEMETRY COLLECTION
- AI-BASED ANOMALY DETECTION
- GPU HEALTH SCORE GENERATION
- PREDICTIVE MAINTENANCE RECOMMENDATIONS
- HISTORICAL TELEMETRY ANALYTICS
- INTELLIGENT ALERT & NOTIFICATION SYSTEM

SYSTEM ARCHITECTURE



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Enterprise AIOps Platform for Intelligent GPU Infrastructure Management*

Technologies and frameworks used for developing the Enterprise AIOps Platform.

AI & Machine Learning

Python
Sclkit-learn
TensorFlow
PyTorch
NumPy
Pandas

GPU Monitoring & Telemetry

NVIDIA
NVIDIA NVML
NVIDIA DCGM
NVIDIA CUDA Toolkit
nvidia-smi

Backend Development

FastAPI
Python
REST API

Frontend Development

React.js
HTML
CSS
JavaScript
Tailwind CSS

Database & Storage

PostgreSQL
TimescaleDB
InfluxDB (optional)

Data Visualization

Plotly
Chart.js
Grafana

Deployment & DevOps

Docker
Git
GitHub
Linux

Development Tools

Visual Studio Code
Jupyter Notebook
Postman

FUTURE INNOVATIONS AND ROADMAP

BUILDING THE NEXT GENERATION OF AI INFRASTRUCTURE MANAGEMENT

- Autonomous Workload Migration – Automatically redistribute AI workloads from high-risk GPUs to healthier GPUs to minimize downtime and maximize cluster efficiency.
- Digital Twin for GPU Infrastructure – Develop a virtual replica of the GPU cluster for real-time monitoring, simulation, and predictive maintenance.
- Reinforcement Learning-Based Resource Optimization – Implement Reinforcement Learning algorithms to optimize GPU scheduling, power consumption, and workload distribution.
- Multi-Data Center Monitoring – Extend the platform to monitor and manage GPU clusters distributed across multiple cloud and on-premise data centers.
- Self-Healing Infrastructure – Enable automatic corrective actions such as restarting GPU services, reallocating workloads, or isolating faulty nodes without human intervention.

OUR VISION IS TO EVOLVE SENTINEL FROM AN INTELLIGENT GPU MONITORING PLATFORM INTO A FULLY AUTONOMOUS AI OPERATIONS (AIOPS) ECOSYSTEM CAPABLE OF SELF-MONITORING, SELF-OPTIMIZING, AND SELF-HEALING NEXT-GENERATION AI INFRASTRUCTURE

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